

PHASE 7-PROJECT DOCUMENTATION

Project Title: POCKETSMART AI

Your Smart Budget & Recommendation Assistant

College Name: Government Arts and Science College, Manur.

College Code: MSU142

Team ID: 6ab21524d329d50f0db6120b

Team Leader: Esakki Aruna S NM ID:55055706AAD495BADB04C8C4D3B478B8

Team Member: Utchi Makali NM ID: 0BBEDB52C642EC0B0BA4CB4B22DE9E2D

Team Member: Kombu Madathi NM ID: 86EE18ABFB36AEF83784FA01234C5948

Team Member: S Ulagammal NM ID: B59DED29E07C4F78BCF442535560F984

1.Introduction

PocketSmart AI is a smart personal budgeting and recommendation assistant designed to help users understand, organize, and manage their day-to-day finances. The system combines expense tracking, budget planning, spending analysis, and personalized recommendations in a single application.

The project focuses on making personal financial management simple and accessible. Instead of requiring users to manually analyze large amounts of transaction information, PocketSmart AI summarizes spending patterns and provides practical recommendations based on the user's budget, expenses, financial goals, and selected preferences.

The proposed application is intended as an educational software project and does not replace professional financial advice. Its recommendations are designed to support better budgeting and informed everyday spending decisions.

2. Problem Statement

Many users find it difficult to maintain a consistent record of their daily expenses and understand where their money is being spent. Traditional budgeting methods often depend on notebooks, spreadsheets, or separate applications for different activities. This can make expense tracking time-consuming and reduce the likelihood of maintaining an accurate budget.

There is therefore a need for a simple assistant that can record expenses, categorize spending, compare actual spending with planned budgets, identify common spending patterns, and provide understandable recommendations.

3. Existing System

Existing personal finance methods commonly include manual notebooks, spreadsheets, calculator-based budgeting, and basic expense-tracking applications. These approaches can record transactions effectively, but they may provide limited personalized analysis or recommendations.

Common limitations include manual data entry, limited visualization, lack of intelligent recommendations, difficulty identifying spending trends, and separation of budgeting and recommendation functions.

4. Proposed System

PocketSmart AI integrates budgeting, expense tracking, analytics, and recommendation features into one system. Users can create a budget, record expenses, view category-wise spending, monitor remaining budget, and receive recommendations generated from their available data.

The system can use rule-based logic and/or a machine-learning recommendation component depending on the implementation. The recommendation engine can consider factors such as spending category, frequency of expenses, budget utilization, savings targets, and user preferences.

5. Objectives

- To provide a simple platform for recording and categorizing expenses.
- To allow users to define budgets for different spending categories.
- To display spending summaries and budget utilization clearly.

- To identify spending patterns from recorded financial data.
- To generate personalized and understandable budgeting recommendations.
- To help users monitor savings goals and reduce unnecessary spending.
- To provide a user-friendly interface for everyday budget management.

6. Scope of the Project

The project covers personal expense management, category-based budgeting, spending analysis, savings goal tracking, and recommendation generation. The initial version can support manually entered expenses and may later be extended to integrate secure financial data sources where legally and technically appropriate.

The project does not aim to execute financial transactions, provide regulated financial advice, approve loans, or make investment decisions on behalf of users.

7. Key Features

- User registration and login.
- Personal profile and budget setup.
- Expense and income entry.
- Expense categorization such as Food, Transport, Education, Shopping, Bills, and Entertainment.
- Monthly and category-wise budget management.
- Dashboard showing income, expenses, remaining budget, and savings progress.
- Spending charts and summaries.
- AI-assisted budgeting recommendations.
- Savings goal tracking.
- Recommendation history.
- Basic alerts for high budget utilization.
- Data export or report generation, if implemented.

8. System Requirements

8.1 Hardware Requirements

- Processor: Dual-core processor or above
- RAM: Minimum 4 GB
- Storage: Minimum 2 GB free space for development and application data
- Display: 1366 × 768 resolution or above
- Internet connection: Required for cloud/AI services if used

8.2 Software Requirements

- Operating System: Windows, Linux, or macOS
- Frontend: HTML, CSS, JavaScript or a suitable frontend framework
- Backend: Python/Node.js/Java or another selected backend technology
- Database: MySQL, PostgreSQL, SQLite, or MongoDB
- AI/ML: Python with suitable data-processing and machine-learning libraries, if applicable
- Development Environment: Visual Studio Code or equivalent IDE
- Browser: Chrome, Edge, Firefox, or equivalent modern browser

9. Functional Requirements

ID	Requirement	Description
FR1	User Authentication	The system shall allow users to register, log in, and securely access their personal budget data.
FR2	Budget Management	The system shall allow users to create, update, and view budgets.
FR3	Expense Management	The system shall allow users to add, edit, delete, and categorize expenses.
FR4	Income Management	The system shall allow users to record income sources.
FR5	Analytics	The system shall calculate totals and display category-wise

		and period-wise spending.
FR6	Recommendations	The system shall generate recommendations based on available user spending and budget information.
FR7	Goals	The system shall allow users to create and monitor savings goals.
FR8	Notifications	The system may notify users when spending approaches or exceeds a selected budget limit.

10. Non-Functional Requirements

- Usability: The interface should be simple and understandable for first-time users.
- Performance: Dashboard and common operations should respond quickly for normal workloads.
- Security: User credentials and financial records should be protected using appropriate authentication and access controls.
- Reliability: Data should be stored consistently and recoverable through suitable backup mechanisms.
- Scalability: The system should allow additional users, categories, and recommendation features to be added.
- Maintainability: The application should use modular components so that individual features can be updated independently.

11. System Architecture

The proposed architecture follows a layered approach:

- Presentation Layer – Provides the web/mobile user interface, forms, dashboard, charts, and recommendation views.

- Application Layer – Handles authentication, budget calculations, expense processing, goal tracking, and business rules.
- AI/Recommendation Layer – Processes relevant user data and generates personalized budgeting suggestions.
- Data Layer – Stores user profiles, income, expenses, budgets, goals, categories, and recommendation records.

Typical data flow: User → Interface → Application/API → Database and Recommendation Engine → Processed Results → Dashboard/Recommendations.

12. Module Description

12.1 User Management

Handles registration, login, profile information, and access control.

12.2 Budget Management

Allows users to set monthly or category-wise spending limits and monitor utilization.

12.3 Expense Tracking

Records transaction date, amount, category, description, and optional payment method.

12.4 Income Tracking

Records salary, allowance, freelance income, or other user-defined income sources.

12.5 Analytics Dashboard

Displays total income, total expenses, remaining budget, category distribution, and trends.

12.6 Recommendation Engine

Analyzes budget and spending information to produce relevant suggestions.

12.7 Savings Goal Module

Allows users to define a target amount and monitor progress toward the goal.

12.8 Reports

Generates summaries that help users review their financial activity for a selected period.

13. Database Design

A relational database design may include the following major entities:

Entity	Important Fields
Users	user_id, name, email, password_hash, created_at
Categories	category_id, category_name, category_type
Transactions	transaction_id, user_id, category_id, amount, transaction_type, date, description
Budgets	budget_id, user_id, category_id, limit_amount, period, start_date, end_date
Goals	goal_id, user_id, goal_name, target_amount, current_amount, target_date
Recommendations	recommendation_id, user_id, recommendation_text, category, created_at

14. AI Recommendation Approach

The recommendation component can initially use explainable rule-based logic and later be enhanced with machine-learning techniques. A rule-based version is easier to test and understand during the first implementation.

Example recommendation logic: if a category's actual spending reaches a high percentage of its allocated budget, the system can display a warning and suggest reviewing recent expenses. If recurring spending is consistently high relative to the user's planned budget, the system can suggest setting a revised category limit or identifying non-essential expenses.

For an advanced version, historical transactions can be transformed into features such as monthly category totals, spending frequency, budget utilization

percentage, and changes over time. A recommendation model can then rank relevant suggestions according to the user's current context. Recommendations should remain transparent and should not claim certainty about future financial outcomes.

15. System Workflow

1. User creates an account and logs in.
2. User enters income and creates a monthly/category budget.
3. User records daily expenses.
4. The application validates and stores transaction data.
5. The analytics module calculates spending totals and budget utilization.
6. The recommendation module evaluates relevant patterns and rules.
7. Personalized suggestions and alerts are shown on the dashboard.
8. User reviews the recommendations and updates the budget or goals as needed.

16. User Interface Description

- Login/Register Screen – Secure entry point for users.
- Dashboard – Provides an overview of income, expenses, budget balance, savings goals, and recommendations.
- Add Expense Screen – Simple form for entering expense details.
- Budget Screen – Shows category limits and utilization.
- Analytics Screen – Presents charts and summaries.
- Recommendations Screen – Displays AI-assisted suggestions with short explanations.
- Goals Screen – Shows progress toward savings targets.

17. Testing Strategy

Testing will be performed at multiple levels to verify correctness, usability, security, and reliability.

Testing Type	Purpose
Unit Testing	Test individual functions such as budget calculation, expense categorization, and recommendation rules.
Integration Testing	Verify communication between the interface, backend, database, and recommendation component.
System Testing	Test complete user workflows from login through expense entry and recommendation display.
Usability Testing	Check whether users can complete common tasks without confusion.
Security Testing	Check authentication, authorization, input validation, and protection of sensitive data.

18. Security and Privacy

- Passwords should be stored using secure password hashing rather than plain text.
- User data should be accessible only to the authorized account.
- Input validation should be applied to prevent invalid or malicious data.
- Sensitive information should be transmitted using secure connections where applicable.
- Only the minimum data required for budgeting and recommendations should be collected.
- Users should be informed about how their data is stored and used.

19. Advantages

- Combines budgeting and expense tracking in one application.
- Provides clear summaries of spending behavior.
- Reduces the effort required for manual budget analysis.
- Provides personalized recommendations based on recorded information.
- Supports savings goal monitoring.
- Can be expanded with additional AI and analytics features.

20. Limitations

- Recommendations depend on the accuracy and completeness of user-entered data.
- The initial system may not support automatic bank transaction synchronization.
- AI recommendations may be limited by the amount of historical data available.
- The system is intended for budgeting assistance and not professional financial advice.
- Privacy and security requirements become more complex if external financial accounts are integrated.

21. Future Enhancements

- Integration with secure financial data providers where available and appropriate.
- Natural-language chat interface for budget-related questions.
- Advanced spending anomaly detection.
- Personalized monthly budget forecasting.
- Receipt scanning and automatic expense extraction.
- Multi-language support.
- Improved visualization and downloadable financial reports.
- Optional family/shared budgeting features with appropriate permissions.

22. Conclusion

PocketSmart AI is proposed as a practical personal budgeting and recommendation assistant that combines expense tracking, budget management, analytics, savings goals, and intelligent suggestions. By presenting financial information in a simple and organized manner, the system can help users understand their spending and make more informed everyday budgeting decisions.

The project provides a foundation for further development in areas such as conversational assistance, automated expense extraction, advanced analytics, and secure financial-data integration.

23. References

1. Python Documentation – <https://docs.python.org/>
2. MySQL Documentation – <https://dev.mysql.com/doc/>
3. MDN Web Docs – <https://developer.mozilla.org/>
4. OWASP Foundation – <https://owasp.org/>
5. scikit-learn Documentation – <https://scikit-learn.org/>